

## **OFFICIAL AUTHORIZATION RECORD**

**Authorization No: PMA/IN-SPACE/AUTH/2026/115**

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Issued Authority: IN-SPACE (Department of Space, Govt. of India).

Authorized Entity: Blue Blocks Montessori Educational Society.

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## Authorization Certificate

For

**Establishment and operations of a hosted  
payload, namely Students of BlueBlocks-1  
(SBB-1)**

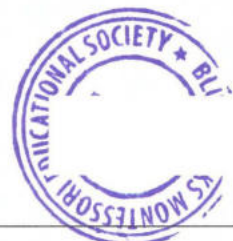


**Indian National Space Promotion and  
Authorization Centre (IN-SPACe), Ahmedabad**

and

**Blue Blocks Montessori Educational Society,  
K.V. Rangareddy, Telangana**

**Date: Jan 07, 2026**



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**AUTHORIZATION CERTIFICATE**

**[See The Gazette of India Notification dated 02<sup>nd</sup> of October, 2021 and Resolution of the Department of Space dated 06<sup>th</sup> of September, 2021]**

**Background:** Reference is invited to the Application submitted by you, Blue Blocks Montessori Educational Society, K.V. Rangareddy, Telangana (hereinafter referred to as 'the Applicant'), seeking Authorization from IN-SPACe for conducting the space activity as set out in SCHEDULE-A (hereinafter referred to as 'Scheduled Space Activity') hereinbelow. As IN-SPACe being the nodal agency of the Government of India responsible for, inter alia, regulating Space Activities and granting Authorization for conducting the Space Activity, we have reviewed your Application and hereby provide Authorization for conducting the Scheduled Space Activity as per SCHEDULE-A, and the terms set out in the **Exhibit-1** including its **Annexure – 1** (Audit Parameters) and technical parameters in **Exhibit-2**.

**SCHEDULE-A****Authorization for Establishment and operations of a hosted payload**

|                                                       |   |                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Nature of the Activity</b>                      | : | Establishment and operations of a hosted payload, namely Students of BlueBlocks -1 (SBB-1) onboard My Orbital Infrastructure-1 (MOI-1) satellite of M/s TM2Space Technologies Private Limited, K.V. Rangareddy (TM2Space).                                                                                                                         |
| <b>2. Authorized Hosted Payload</b>                   | : | Hosted Payload: SBB-1<br>Host Space Object: MOI-1 Satellite<br>Host Entity: TM2Space                                                                                                                                                                                                                                                               |
| <b>3. Application No</b>                              | : | a) IDP Application no. dt. 26 Jun 2025;<br>b) Subsequent information/details furnished to IN-SPACe vide IDP and emails dated 28 Aug 2025, 29 Aug 2025, 18 Sep 2025, and any other information provided by the Applicant to IN-SPACe during the course of processing of the authorization.<br>[a) and b) together referred to as the 'Application'] |
| <b>4. Authorized Entity and its Registered Office</b> | : | Blue Blocks Montessori Educational Society, K.V. Rangareddy, Telangana (BBMES)<br>BBMES is an educational society established on 01 Apr. 2016 and is registered under the Telangana Societies Registration Act, 2001 having registration no. as 620/2016.<br>The registered office address is:                                                     |





|                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>5. Validity Period</b>                                                                                                        | : | <p>a) Validity of this Authorization is:<br/>either one (1) year from the date of launch (declared operational life of SBB-1),<br/>or<br/>end of life of Hosted Payload or Host Space Object,<br/>or<br/>validity period of the MoU signed between Hosted Payload and Host Entity,<br/>whichever is earlier ('Validity Period').</p> <p>a) SBB-1 must be launched and operationalize, within three months of the date of issuance of this Authorization Certificate, failing which this Authorization shall stand null and void unless it is amended for the new launch period.</p> <p>The Applicant shall inform the date of launch and operationalization of SBB-1 to IN-SPACE in writing and through e-mail within seven (7) working days.</p> <p>b) The Applicant shall seek separate Authorization from IN-SPACE to conduct the Space Activity beyond this Validity Period.</p> |
| <b>6. Details of Authorizations and approvals obtained from other agencies, departments and the ministries of the Government</b> | : | <p>ITU filing is done by Host Entity.</p> <p>New ITU Filing – MOI-1, API receipt date on ITU website is 28 Nov. 2025. The Filing process is completed through WPC and further details are available on ITU Database with Notice ID / Reference:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>7. Objective/ Intended Use</b>                                                                                                | : | <p>Objectives of SBB-1 is to monitor the environment and orientation in Low Earth Orbit using compact, reliable, and space-suitable sensors, namely BME280, BNO055, and PT100 RTD. It is totally dependent on the host space object (MOI-1) for communication channel, MCC and TT&amp;C operations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



*Disclaimer: This authorization is being provided based on the information, declaration and confirmation provided by the Applicant on various aspects/considerations in its Application to IN-SPACe and subsequent additional information/clarification provided during the course of processing of the Application. This authorization does not guarantee or endorse the technical performance, compliance to the specifications, or quality of service by the Applicant.*

**Director, PMA, IN-SPACe**

**डॉ. प्रफुल्ल कुमार जैन / Dr. Prafulla Kumar Jain**

निदेशक / Director

कार्यक्रम प्रबंधन एवं प्राधिकरण निदेशालय /

Programme Management and Authorization Directorate

भारतीय राष्ट्रीय अंतरिक्ष संवर्धन एवं प्राधिकरण केंद्र /

Indian National Space Promotion and Authorization Centre

( इन-स्पेस / IN-SPACe)

अंतरिक्ष विभाग, भारत सरकार / Dept. of Space, Govt. of India

अहमदाबाद/Ahmedabad-380058, गुज., भारत / Guj., India



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**DECLARATION**

By counter signing this Authorization Certificate, the Applicant hereby acknowledges that it shall be legally obligated to comply with, abide by and obey the terms and regulations set out in this Authorization Certificate including SCHEDULE-A, 'Terms of Authorization' set out in the Exhibit-1 including its Annexure-1 and 'Technical Parameters' in Exhibit-2.

AGREED TO AND ACCEPTED THE TERMS OF THE AUTHORIZATION



[Signature with Seal of the authorized signatory of the Applicant]

Name : PAVANI GOYAL  
Designation : FOUNDER PRESIDENT  
Mobile No. :  
Address :  
  
Place :  
Date :





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EXHIBIT-1

TERMS OF AUTHORIZATION

1. Representations and Warranties:

- a. The Applicant represents and warrants that all the information provided by the Applicant as part of the Application and process for securing Authorization is true and correct.
- b. The Applicant acknowledges that space remains a domain which has implications for national interests and national security. Therefore, if any of the information provided in the Application or otherwise is found to be, or becomes incorrect or false at any stage, IN-SPACe reserves the right to, inter alia,
  - i. Withdraw the Authorization from the Applicant.
  - ii. Exercise any or all remedies available under law to recover the losses suffered by IN-SPACe or the Government of India and claim damages caused directly or indirectly from the Applicant, as a result of a breach of sub clause (a) above.

Provided that no decision shall be taken in this regard by IN-SPACe without providing an opportunity of hearing to the Applicant.

- c. The Applicant represents and warrants that it has the power and authority to execute this Authorization Certificate and to perform its obligations thereunder and such execution and performance of its obligations will not breach its memorandum or articles of association, or constitute default under or violation of applicable laws or other obligations to which it is bound or any order, judgment or decree of any judicial, quasi-judicial or government body.
- d. The Applicant represents and warrants that it has full legal power and authority and all licenses, permits, and authorizations necessary to carry on the businesses in which it presently is engaged and in which it proposes to engage in, pursuant to this Authorization Certificate.
- e. The Applicant represents and warrants that the Applicant, its controlling shareholders, promoters, directors or officers, are not subject to any legal, quasi-legal, administrative, arbitration, mediation, conciliation or other proceedings, claims, actions, governmental investigations, orders, injunctions, judgments or decrees of any nature, existing or threatened which may prejudicially affect its ability to carry out the Scheduled Space Activity specified in this Authorization Certificate.
- f. The Applicant represents and warrants that it shall comply with the Norms, Guidelines and Procedures for implementation of the Indian Space Policy 2023 ('ISP-2023') in respect of Authorization of Space Activities ('NGP') issued by IN-SPACe (presently Version 1.0 Document no. IN:ISP2023:NGP2024/V1.0 dated May 03, 2024), and its revisions or other guidelines/directives issued from time to time.





- g. The Applicant represents and warrants that it meets the specified criteria, included but not limited to financial solvency, legal compliance, technical capability, and managerial capacity, amongst others.
- h. The Applicant represents and warrants that it is in compliance with the existing laws, rules, regulations, and policies. The Applicant commits to remain compliant with any amendments made to the extant new applicable laws, rules, regulations, and policies as and when they are notified, and with any new and applicable laws, rules, regulations, and policies. The Applicant further assures the accuracy and completeness of all the documentation and information submitted for consideration.
- i. The Applicant represents and warrants that it shall take all possible measures for (a) limiting the generation of space debris in accordance with the applicable Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space (UN-COPUOS) (2007) and their future revisions if any, along with the Norms Guidelines and Procedures for 'Registration of the Space Object' provided in the Chapter-XI of the NGP, and the further revisions, if any, and (b) limiting adverse environmental effects on Earth or in outer space or adverse changes in the atmosphere.

## 2. Scope of this Authorization:

- a. The description and nature of the authorized Space Activity is set out in SCHEDULE-A of this Authorization Certificate (hereinafter referred to as the Scheduled Space Activity). The Authorization is confined only to the Scheduled Space Activity described in the SCHEDULE-A and does not authorize any Space Activity which is not explicitly included in the SCHEDULE-A of the Authorization Certificate.
- b. In the Authorization Certificate, except for the Applicant and as provided in sub clause (d) of clause 2 herein, no other person or entity including the subsidiaries of the Applicant, holding companies of the Applicant, or persons controlling the Applicant, controlled by the Applicant or under the control of the same person(s) who control the Applicant, either by virtue of shareholding or contract, can conduct the Scheduled Space Activity set out in the Authorization Certificate.
- c. The Applicant shall inform IN-SPACe in writing and through email, within 48 (forty-eight) hours of any change in its 'Management and Control' or in its partnership or trust, as applicable, or shareholding pattern. IN-SPACe in its sole discretion or in consultation with the relevant departments/ministries wherever necessary, may decide to revoke, amend or issue fresh Authorization Certificate in such cases.

The Applicant shall seek a fresh Authorization in case of any change in the Management and Control of the Applicant company or an equivalent change in



the partnership or trust, resulting in transfer of the 'Control' to non-Indian Entity/ Entities or if the Applicant is no longer under Indian Management and Control.

- d. The Authorization, so granted, is not transferrable to any third parties. ("prospective transferees"), except with the prior written approval of a duly empowered officer of IN-SPACE, which IN-SPACE in its sole discretion may grant. Provided, however, that such prior written approval shall not be granted, unless:
- i. The prospective transferees have the same or better technical, financial and other capabilities, prescribed by IN-SPACE to conduct the Scheduled Space Activity prescribed in the Authorization Certificate.
  - ii. The prospective transferees subscribe to and undertake to comply with the terms set out in the Authorization Certificate and the terms set out herein and all other applicable domestic and international law.
  - iii. The Applicant executes an unconditional performance guarantee for a term to be set out by IN-SPACE, acting as a surety for the compliance of and performance of the obligations set out in these terms and other applicable, laws, rules and regulations by the prospective transferee.
- e. Provided further that in case of IN-SPACE approving in writing, the transfer of any Authorization pursuant to sub clause (d) of clause 2 herein, it shall reserve the right to a) impose such conditions on the transfer and the prospective transferee as well as the Applicant and/or b) suspend, withdraw or terminate any benefits or exemptions or waivers provided to the Applicant.

### 3. Effect of this Authorization:

In the event of any conflict between the terms and conditions set out in this Authorization, and any agreement executed by the Applicant, including but not limited to, the agreement for launching an object into space, license agreements with the DoT for the grant of frequency and/or operational license to deploy services, the terms of this Authorization shall prevail, in so far as the subject matter herein is concerned.

### 4. Terms of Authorization, in addition to and not in derogation of other laws:

The terms of the Authorization set out herein are in addition to and not in derogation of other laws or license terms or conditions applicable to the Applicant or the Applicant's Space Activities, including but not limited to rules, guidelines, notifications, circulars and regulations of the Department of Science and Technology, Ministry of Home Affairs, Ministry of Defense, Department of Telecommunications, Telecom Regulatory Authority of India and any other authority satisfying the definition of "State" under Article 12 of the Constitution of India.





**5. Powers of Amendment:**

IN-SPACE reserves the right to amend or modify including add, substitute or repeal the terms of the Authorization, provided however that IN-SPACE will,

- a. Not retrospectively amend the terms of Authorization so as to render the Applicant ineligible for Authorization or for withdrawal of Authorization, unless it is, in IN-SPACE's discretion necessary in national interests to do so.
- b. Not amend the terms of the Authorization unless it is, in IN-SPACE's determination beneficial for national interests and national security.

All such amendments to the terms of the Authorization shall, when communicated privately to the Applicant or when published on the official website of IN-SPACE, as the case maybe, deemed to be part and parcel of the Authorization granted to the Applicant.

**6. Specific Regulations and Terms of Authorization:**

- a. The operations of the SBB-1 Hosted Payload shall be in accordance with Host Space Object ITU network filing namely MOI-1, made by the Host Entity TM2Space and the assignment of the frequencies to the Host Entity by WPC Wing of DoT.
- b. The Applicant shall be in full control of all the operations of the SBB-1 Hosted Payload and shall ensure that its mission objective, including intended users, is not of concern from the national security perspective.
- c. The Applicant shall use only IN-SPACE Authorized MCC of the Host Entity for the operations of SBB-1 Hosted Payload. IN-SPACE has issued the authorization for the Host Entity's self-owned MCC located at IIIT-Hyderabad, Telangana vide Authorization Ref. PMA/IN-SPACE/AUTH/2025/108 dated 17 Nov. 2025.
- d. The Applicant shall comply with the NGP (presently Version 1.0 Document No: IN:INP2023:NGP2024/V1.0 dated May 03, 2024), and its revision or other guidelines/directives issued from time to time.
- e. The Applicant shall ensure, through Host Entity that appropriate arrangements are in place in respect of cyber security and information security in the operations of MCC facility in order to ensure that no unauthorized access is made to the data being transmitted/ received to/from the SBB-1 Hosted Payload.
- f. The Applicant, through Host Entity, at all times, shall make available, the data being received/ transmitted from the TT&C and MCC stations, to IN-SPACE or other departments/agencies of Government of India, as and when demanded and without undue delay, for the reasons involving national security or such critical considerations.



- g. The Applicant, through Host Entity, shall employ appropriate measures in respect of the safety and security including cyber security aspects during the operations of the TT&C and MCC stations.
- h. Any change in the location of the MCC shall require approval of IN-SPACe.
- i. The data reception earth stations for acquisition of the Hosted Payload data by the Host Entity are presently located outside Indian Territory. IN-SPACe may seek the details of such stations, if necessary, to assess whether the location of such stations outside the territorial jurisdiction of India does not pose a threat to the national security, foreign relations, intelligence and security operations in the interests of the Republic of India.
- j. The Authorization shall be applicable for the SBB-1 Hosted Payload which is being authorized and shall be valid till the end of the declared operational life of the Hosted Payload. Any change of the payload configuration of SBB-1 shall require a fresh Authorization. Further, the Applicant may seek the extension of the operational life and the extension of the validity of the Authorization of the SBB-1 Hosted Payload, maximum up to end of operational life of the Host Space Object or Validity Period of the MoU signed between Hosted Payload owner and Host Entity – whichever is earlier, subject to the assessment of the SBB-1 Hosted Payload and MOI-1 Host Space Object health parameters and space situational awareness aspects by IN-SPACe, and the same shall be granted by IN-SPACe at its discretion.
- k. IN-SPACe may impose control on operations of the SBB-1 Hosted Payload and Scheduled Space Activity or terminate/suspend the Authorization, during national emergencies or in the interest of national security, such as situation arising out of or due to conflicts or natural disasters or times of emergency as declared or specified by Government of India or other such contingencies as may be specified by IN-SPACe or the Government of India. The Applicant may be given a prior notice of such measures/termination or suspension, wherever possible.
- l. The Applicant shall notify IN-SPACe in writing and through e-mail within 30 (thirty) days of any event or factor that could affect the validity or the continuance of the granted Authorization or of any change with regard to the meeting of conditions under which the Authorization was granted.
- m. The objectives and operations of the SBB-1 Hosted Payload shall be in compliance with the obligations under applicable international law, and international space-related treaties, agreements, guidelines, principles or standards, such as the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies ("Outer Space Treaty"); Convention on International Liability for Damage Caused by Space Objects ("Liability Convention"); Convention on Registration of Objects Launched into Outer Space ("Registration Convention"), etc.





- n. For the reasons involving national security, geopolitical considerations or harmful interference to other satellite networks, IN-SPACe in consultation with other departments/ministries, may regulate the operation of the SBB-1 and the Applicant shall cease the emissions and operations of the SBB-1, without any delay, in such cases.

#### 7. Obligations of the Applicant:

While conducting the Scheduled Space Activity pursuant to the Authorization granted, the Applicant will be bound by the following obligations:

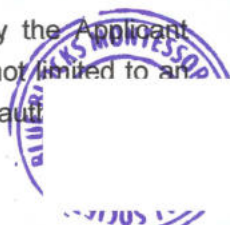
- a. The Applicant will sustain and maintain the technical and financial abilities as demonstrated at the time of receiving the Authorization for conducting the Scheduled Space Activity, throughout the duration of the Scheduled Space Activity within the Validity Period. For the sake of clarity and for purposes of illustration only, it is clarified as follows:
- i. Explanation a: The duration of the Scheduled Space Activity commences from the time an object is made fit for integration into the vehicle launching it into space, until the time that it is safely disposed of in accordance with orbital debris remediation measures promulgated by the state from time to time.
  - ii. Explanation b: For the proposed downstream Scheduled Space Activities, the time from when the infrastructure necessary to uplink and downlink or transmit data to and/or from the object in space is established, until the time its technical abilities to uplink and downlink or impact or affect the Space Object ceases, shall be considered as the duration of the Scheduled Space Activity.
- b. The Applicant, if so wishes, shall seek a separate extension of the Authorization Validity Period from IN-SPACe to conduct the Scheduled Space Activity beyond the duration of the Scheduled Space Activity authorized by IN-SPACe under this Authorization Certificate. Unless the permission for the extension of the duration of the Scheduled Space Activity is provided separately by IN-SPACe, in writing, the Applicant must not presume that the Authorization is deemed to be extended. Applications for extension of the duration of the Scheduled Space Activity originally authorised must be filed one (1) month prior to expiry of the term or duration of the Scheduled Space Activity originally authorized. However, if such extension of the Authorization is not granted by IN-SPACe in spite of the expiry of one (1) month of the receipt of the Application for such extension, the Authorization shall be deemed to be extended until such time IN-SPACe conveys its decision on the Authorization for the extended duration of the Scheduled Space Activity. While calculating the time limit of one (1) month for IN-SPACe to make its decision, any time taken by the





Applicant to furnish information or clarification pursuant to a written query from IN-SPACe, shall be excluded.

- c. The Applicant shall conduct the Scheduled Space Activity in a transparent manner with due regard to the rights of other person(s) and states for conducting Space Activities and in the interest of maintaining international peace and security and promoting international co-operation and understanding. The Applicant shall ensure that the Scheduled Space Activity do not cause any damage to the surface of the earth, any aircraft, other Space Object or to any property or person.
- d. The Applicant shall conform to the Standard Operating Procedures, Norms, Guidelines and Procedures for registration of a Space Object brought out by IN-SPACe and shall not do any act, deed or thing to violate such registration or violate the conditions of such registration.
- e. The Applicant shall conform to the framework for safe and sustainable space operations as and when brought out by the Department of Space.
- f. The Applicant shall comply with, shall assist and enable the Government of India in complying with its obligations under, and shall not cause the Government of India to be breach of, or incur any liability, under, applicable international law, and applicable international space-related treaties, agreements, guidelines, principles or standards ("**International Treaties**") to which India is a signatory, or becomes a signatory during the term of this Authorization, or adopted by India, including but not limited to:
  - i. The Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies ("Outer Space Treaty");
  - ii. Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched into Outer Space ("Rescue Treaty");
  - iii. Convention on International Liability for Damage Caused by Space Objects ("Liability Convention");
  - iv. Convention on Registration of Objects Launched into Outer Space ("Registration Convention");
  - v. Agreement Governing the Activities of States on the Moon and Other Celestial Bodies ("Moon Agreement");
  - vi. Artemis Accords;
  - vii. Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space.
- g. Where any act, omission, information received or discovery by the Applicant places an obligation on the Government of India, including but not limited to an obligation to register a Space Object, or to notify an international authority



international law or International Treaties, the Applicant shall notify IN-SPACE immediately, and no later than 24 (twenty-four) hours after such act, omission, intimation or discovery.

- h. Where any act or omission of the Applicant exposes the Government of India to any liability under international law or International Treaties, IN-SPACE or the Government of India shall be entitled to issue directions to the Applicant for providing mitigation measures or assistance, financial or otherwise, in accordance with NGP issued by IN-SPACE, from time to time, in furtherance of the ISP-2023 or otherwise, and the Applicant shall comply with the same. This is without prejudice to any claim that the Government of India or IN-SPACE may make against the Applicant.
- i. The Applicant acknowledges its responsibility to remain updated on any additions, changes or amendments to applicable international law and International Treaties during the term of this Authorization Certificate, and agrees to comply therewith.
- j. The Applicant shall ensure compliance with any domestic legislations, rules, regulations or policies formulated, from time to time, by the Government of India to comply with applicable international law and International Treaties.
- k. During the Validity Period of this Authorization, the Applicant shall provide information to IN-SPACE, in writing as well as by email, regarding filing of any criminal complaint or report, including first information report, registration of criminal case(s), allegations, chargesheets, trials, conviction by courts, etc., if any, against the Applicant, its controlling shareholders or promoters or beneficial owners or directors or officers or partners or trustees of the Applicant company for the offences under Chapters VI, VII, VIII, IX and X of the Bhartiya Nyaya Sanhita, Chapters VI, VII, VIII, IX, IXA, XIV, XVI and XXII of the Indian Penal Code, 1860, offences under the Official Secrets Act, 1923, offences under the Prevention of Corruption Act, 1988 and/or offences under the Prevention of Money Laundering Act, 2002 and/or aiding and/or abetting, or engaging in a conspiracy for the commission of any of the foregoing offences and/or such other offences that may be considered serious offences or as may be notified by IN-SPACE from time to time in writing.

Change in the status of the declaration made by the Applicant if any in the application or during the processing of IN-SPACE Authorization in respect of such offences as stated above, shall be promptly updated and informed to IN-SPACE by the Applicant in writing as well as by email during the Validity Period of this IN-SPACE Authorization.

IN-SPACE, taking note of such offences or status provided by the Applicant, reserves the right to terminate/suspend/revoke the Authorization. The Applicant may be given prior notice of such termination/suspension, wherever possible.





- l. The Applicant shall cooperate with IN-SPACE, the Government of India and its agencies in demonstrating compliance with international law and International Treaties, including transparency and confidence building measures.
- m. The Applicant shall promptly notify IN-SPACE of any changes in its operations or services that may affect its compliance with international law or International Treaties, through a written notice as well as by email within 24 (twenty-four) hours.
- n. The Applicant desiring to discontinue/terminate/withdraw, partially or fully, the Scheduled Space Activity authorized by IN-SPACE anytime during the term of the Authorization granted by IN-SPACE shall provide a prior notice of at least 30 (thirty) days, in writing and through e-mail, to IN-SPACE giving reason/s for such discontinuation/termination/withdrawal.

In case of such discontinuance, withdrawal or termination, the Applicant shall provide appropriate assistance to IN-SPACE, including but not limited to support, transition assistance, sharing of data, as may be determined by IN-SPACE in its sole discretion, and such obligation shall be specifically enforceable.

- o. The Applicant shall comply with the plan outlined for the Scheduled Space Activity in its Application for Authorization to IN-SPACE and in its applications before international organizations such as for obtaining orbital slot and/or frequency from the International Telecommunication Union.

Any deviation from the plan, which could materially affect space situational awareness and also the Space Activities of other nations, such deviations shall not be undertaken without the prior written approval of IN-SPACE, which may be provided, if IN-SPACE is satisfied of the following,

- i. The deviation does not pose a threat to the sustainability of space or to the Space Objects of other nations or to lives or property within the air space or in the territory of Earth.
  - ii. The Deviation is unlikely to trigger security concerns among other nations whose Space Activities could be affected by the Applicant's Scheduled Space Activity.
- p. The Applicant shall not carry out any Space Activities or leverage the Authorization pertaining to the Scheduled Space Activity to perpetrate any act or omission that violates any international or applicable domestic law(s).

Further, the applicant shall comply with the geospatial, telecommunication and other relevant guidelines, policies, rules and regulations issued by concerned regulatory authorities under applicable jurisdiction.

- q. The Scheduled Space Activity shall not be conducted in violation or contrary to public health, public policy and morality. For example, if upon a finding of any violation of public health, policy and morality concerns, any other government agency or authority revokes the licenses of the Applicant or raises





IN-SPACE, then IN-SPACE, reserves the right to revoke its Authorization after affording the Applicant an opportunity of being heard.

- r. The Applicant acknowledges that IN-SPACE, in its sole discretion, if it is satisfied that it is necessary and expedient to do so in the interest of national security, friendly relations with foreign states, or in the event of war or emergency can:
  - i. Use the equipment, services, network, and infrastructure of the Applicant used to provide the services authorized by this Authorization ("Applicant Infrastructure");
  - ii. Suspend or prohibit the use of the Applicant Infrastructure from specified countries or persons;
  - iii. Take over the control, management, and operation of the Applicant Infrastructure; and
  - iv. Take any other action that IN-SPACE deems expedient to do so in the interest of national security.
- s. The Applicant shall not carry out any activities which can jeopardize the sovereignty and security of India or friendly or diplomatic relations with foreign states or actors.
- t. The Applicant shall comply with and be governed by NGP issued by IN-SPACE (presently Version 1.0 Document no. IN:ISP2023:NGP2024/V1.0 dated May 03, 2024), and its revisions or other guidelines/directives issued from time to time.

## 8. Material Adverse Events and Change Events

- a. Material Adverse Events for the purposes of this Authorization are those events or incidents which
  - i. Threaten human life, environment or property in the territory of the earth
  - ii. Threatens the security or territorial integrity of the Republic of India
  - iii. Poses a threat to other objects in space.
  - iv. Poses a risk to the public order and national security
  - v. Expose the Republic of India to liability under International Law or International Treaties or to a third party.
- b. In the event of a Material Adverse event occurring in respect of the Scheduled Space Activity, the Applicant shall take immediate measures to mitigate and avoid any damage from such a material adverse event and shall report IN-SPACE in writing and through email about such event within 24 (twenty-four) hours of the occurrence, and also shall report to IN-SPACE within 7(seven) days about the mitigation measures taken.



- c. For any incident or event occurring in respect of a Scheduled Space Activity which does not qualify as a material adverse event but could impact the design, intent and overall nature of the Scheduled Space Activity authorized by IN-SPACE ("Change Event"), the same shall be reported to IN-SPACE within 30 (thirty) days of the occurrence of the Change Event.

#### 9. Liability from Space Activities:

- a. IN-SPACE shall, from time to time, promulgate policies, rules, guidelines and regulations governing the Applicant's liability for conducting Space Activities ("liability guidelines") and the same shall and always be treated as part and parcel of the terms of the Authorization, including any amendments to such liability guidelines.
- b. Applicant shall undertake the establishment and operations of the Hosted Payload, at its own risks and peril, and shall be liable for any third-party damage claims including those covered under the Liability Convention or otherwise arising in consequence to the hosted payload launch, establishment and operations.

The Applicant is advised to undertake appropriate measures such as maintain adequate insurances to cover all such risks and eventualities. However, Applicant may revisit such measures upon intimation by IN-SPACE to the Applicant of the competent authority's approval, if any, exempting the Applicant from maintaining Third Party Liability (TPL) insurance covering claims under the Liability Convention.

- c. However, notwithstanding any liability guidelines, if on an investigation conducted by IN-SPACE, the Applicant is found guilty of willful misconduct or neglect resulting in liability, then IN-SPACE reserves the right to direct the Applicant to pay such damages as are necessary to remedy losses directly arising out of such misconduct or negligence, after providing the Applicant an opportunity of being heard.
- d. Any loss, damage, cost, fees or other expenses incurred by IN-SPACE or the Government of India or its agencies under the Liability Convention or otherwise, which are attributable to the Applicant shall be recoverable from the Applicant in full, along with legal costs and expenses incurred for such recovery, in accordance with NGP issued by IN-SPACE, from time to time, in furtherance of the ISP- 2023 or otherwise.

#### 10. Termination for default or breach of the conditions of Authorization:

IN-SPACE reserves the right to withdraw and/or terminate the Authorization after providing prior written notice of 7 (seven) working days in writing to the Applicant, in the event of a breach of any of the terms, conditions, etc.





warranties of this Authorization or any law, rules, regulations or applicable guidelines issued by IN-SPACE or DOS including but not limited to the guidelines provided in respect of the Authorization of Space Activities including those provided in Document no. IN:ISP2023:NGP2024/V1.0, and other guidelines or in the event of default by the Applicant in carrying out activities under this Authorization Certificate or on occurrence of a Material Adverse Event or Change Event; or in the event of any change in its 'Management and Control' or shareholding pattern or an equivalent change in a partnership or trust, as applicable of the Applicant or in the event that the Applicant violates the guidelines provided in this Authorization Certificate.

Provided however, that if the Applicant is able to rectify or remedy the breach to the satisfaction of IN-SPACE, IN-SPACE reserves the right but not the obligation to restore the Authorization so terminated or revoked.

Provided further that IN-SPACE or the Government of India or its agencies shall not be liable for any damage or loss caused or arisen out of aforesaid termination or withdrawal of Authorization.

**11. Termination of this Authorization during national emergencies or in the interest of national security:**

(1) In specific circumstances, such as

- a. During times of conflict or natural disasters or times of emergency declared in accordance with the constitution of India or law,
- b. Where a criminal complaint or report, including first information report is registered/filed, or a criminal case is registered, or a chargesheet is filed, or a trial is conducted and a conviction is made against the Applicant, its controlling shareholders or promoters or beneficial owners or directors or officers or partners or trustees in respect of offences under Chapters VI, VII, VIII, IX and X of the Bhartiya Nyaya Sanhita, Chapters VI, VII, VIII, IX, IXA, XIV, XVI and XXII of the Indian Penal Code, 1860, offences under the Official Secrets Act, 1923, offences under the Prevention of Corruption Act, 1988 and/or offences under the Prevention of Money Laundering Act, 2002 and/or aiding and/or abetting, or engaging in a conspiracy for the commission of any of the foregoing offences and/or such other offences that may be considered serious offences or as may be notified by IN-SPACE from time to time, in writing, or
- c. Such other contingencies as may be specified by IN-SPACE or otherwise, or determined by IN-SPACE in writing, as being reasons beyond its control or the control of the Government of India (collectively referred to as "Legitimate Withdrawal Contingencies"),

IN-SPACE reserves the right to withdraw this Authorization for conducting the Scheduled Space Activity or exercise, or take the exclusive, sole



control ("control measure/s") over a Space Object or the Scheduled Space Activity, as applicable.

Provided further that IN-SPACE or the Government of India or its agencies shall not be liable for any damage or loss caused or arisen out of aforesaid actions.

(2) While exercising such a right, IN-SPACE will consider:

- a. Providing a reasonable prior notice to the Applicant, wherever possible.
- b. The nature and extent of the Legitimate Withdrawal Contingencies and the necessity for withdrawing the Authorization or imposing control measures to effectively respond to the Legitimate Withdrawal Contingencies.
- c. The period or the extent to which the Authorization so granted ought to be withdrawn or control measures to be imposed, in order to effectively address and respond to the Legitimate Withdrawal Contingencies; and
- d. The nature and extent of the costs and the expenses and investments incurred by the Applicant to act upon the Authorization and whether such losses are irrecoverable in nature.

(3) In case of termination/revocation of the Authorization under clauses (10) or (11), the Applicant shall provide appropriate assistance to IN-SPACE, including but not limited to support, transition assistance, sharing of data, as may be determined by IN-SPACE in its sole discretion, and such obligation shall be specifically enforceable.

## 12. Limitation of Liability:

In the event of any termination or revocation of this Authorization by IN-SPACE, neither IN-SPACE nor the Government of India shall be liable for any loss of profits, revenue, or any indirect or consequential damages incurred by the Applicant, its contractors, subcontractors, or customers as a result of any termination of this Authorization.

## 13. Other rights of IN-SPACE:

IN-SPACE reserves the right to conduct periodic inspections or audits at the Applicant's premises and operations, in line with the Audit parameters included in Annexure-1 of EXHIBIT 1, by itself or through its authorized representative to ensure compliance with the terms of this Authorization Certificate and relevant laws and regulations.





**14. Jurisdiction & Governing Law:**

By applying for and/or acting upon this Authorization, the Applicant agrees to irrevocably and unconditionally submit to the exclusive jurisdiction of the Republic of India and the laws existing in force by the Republic of India concerning this Authorization and the Scheduled Space Activity. This Authorization and any dispute or claim arising out of or in connection with it or its subject matter, existence, negotiation, validity, termination or enforceability (including non-contractual disputes or claims) shall be governed by, and construed in accordance with, the laws of the Republic of India.

**15. Dispute Resolution:**

Any dispute arising out of or in connection with this Authorization or the terms set out herein or conducting of the Scheduled Space Activity pursuant thereto, shall be resolved amicably through mediation and intervention by an independent committee constituted by IN-SPACe Board, as far as possible. If the resolution is still not achieved, Arbitration proceedings under the Arbitration and Conciliation Act, 1996 shall be initiated. Arbitration panel will include three (3) arbitrators, and each party shall appoint one arbitrator having technical experience in the field of Space science and technology. The two appointed arbitrators shall appoint the presiding arbitrator having experience in the field of international law/arbitration. The arbitration shall be conducted by the Gujarat High Court Arbitration Centre, Ahmedabad. The seat of arbitration shall be Ahmedabad, and courts of Ahmedabad shall have exclusive jurisdiction over any disputes.

**16. Notices:**

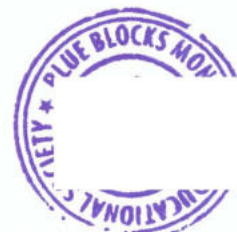
Any notice in connection with the terms of the Authorization shall be provided in writing **as well as through email** to the address mentioned below:

**a. To IN-SPACe:**

Director, Program Management & Authorization (PMA),  
IN-SPACe Headquarters,

**b. To Applicant:**

Mr. Pavan Goyal, President  
Blue Blocks Montessori Educational Society,



**17. Indemnification:**

The Applicant shall defend, indemnify IN-SPACe and Government of India, their respective officers, directors, employees, consultants, affiliates, subsidiaries and agents ("IN-SPACe Indemnitees") harmless against any direct or indirect loss, damage or costs (including attorneys' fees incurred for national and international courts representation) incurred in connection with claims, demands, suits, or proceedings ("Claims") arising out of conducting the Scheduled Space Activity under this Authorization, including by way of (i) breach of terms, conditions, representations and warranties of this Authorization; (ii) breach of any applicable domestic and international law and International Treaties; (iii) violation of any third party rights; (iv) claims of bodily injury to any person or damage to property; or (v) wilful negligence or misconduct of the Applicant.

**18. General:**

The Applicant shall extend all assistance, co-operation and compliance as required by IN-SPACe in the exercise of IN-SPACe and the Government of India's functions under this Authorization.

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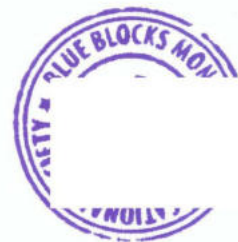




**Annexure 1**  
**Audit Parameters**

| Periodicity of Audit: Once in the operational life of the hosted payload |                                                                                                                       |                                                                                                                                                                                               |                                                                                                       |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sl. No                                                                   | Details                                                                                                               | Factors to be verified                                                                                                                                                                        | Remarks                                                                                               |
| 1.                                                                       | Credentials of Authorized Entity                                                                                      | Declared financial solvency, legal compliance, technical capability, regulatory compliance (existing laws, rules, regulations, and policies) and managerial capacity of the Authorized Entity | To be verified with the information provided in the application form and through latest Annual Report |
|                                                                          |                                                                                                                       | Shareholding pattern of Authorized entity;                                                                                                                                                    |                                                                                                       |
|                                                                          |                                                                                                                       | Management & Control of the Authorized Entity                                                                                                                                                 |                                                                                                       |
|                                                                          |                                                                                                                       | FDI infusion in the Authorized entity and change in its shareholding pattern thereafter; Addition of any new Director/Key Managerial Personnel.                                               |                                                                                                       |
| 2.                                                                       | Hosted payload Mission                                                                                                | Revision in launch date                                                                                                                                                                       |                                                                                                       |
|                                                                          |                                                                                                                       | Verification of Orbital parameters achieved                                                                                                                                                   |                                                                                                       |
|                                                                          |                                                                                                                       | Date of operationalisation of the hosted payload/payload (initiation of the intended services from the hosted payload)                                                                        |                                                                                                       |
|                                                                          |                                                                                                                       | Health of the Hosted Payload (verification of the balance operational life)                                                                                                                   |                                                                                                       |
| 3.                                                                       | Ground Segment                                                                                                        | Confirmation on location of MCC Stations declared in the application                                                                                                                          | Confirmation on using IN-SPACE authorized MCC station                                                 |
|                                                                          |                                                                                                                       | Confirmation on the tele-commanding capability from the MCC in India                                                                                                                          |                                                                                                       |
|                                                                          |                                                                                                                       | Verification of MCC security mechanism                                                                                                                                                        |                                                                                                       |
| 4.                                                                       | Audit of terms & conditions specific to the Authorization Certificate issued to the Authorized Entity, as applicable. | As per T&C                                                                                                                                                                                    |                                                                                                       |

**Note:** Both the Auditor/IN-SPACE and the Applicant shall sign the Audit Report along with an undertaking from the Applicant to the effect that the information provided by the Applicant during the audit is true to the best of its knowledge.



|                                                             |                                                                                                             | <ul style="list-style-type: none"><li>○ Purpose: Sends environmental and orientation data from BME280, BNO055, and PT RTD to the Host system.</li><li>○ Downlink to Ground (via Host Space Object)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------|-------------|---------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------|
| 4.                                                          | Launch Details                                                                                              | Host Space Object: My Orbital Infrastructure – 1 (MOI-1) Satellite<br>Launch vehicle name: PSLV C-62<br>Launch Facility: SDSC, Sriharikota<br>Launch window: December 2025 – January 2026 (tentative)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 5.                                                          | Orbital Parameters                                                                                          | Altitude: ~505 km (tentative and subject to change post launch mission finalization)<br>Inclination: 97.5 degree<br>Near circular Sun Synchronous Polar Orbit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 6.                                                          | Operation life of the payload                                                                               | ~1 year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 7.                                                          | TT&C and Data reception station details                                                                     | SBB-1 Hosted payload is totally dependent on the communication channel and MCC of the Host Entity (TM2Space).<br>Multiple stations for TT&C and Payload Data downlink within and outside India are proposed by Host Entity (TM2Space) through GSaaS operators like: LeafSpace and Dhruva Space.                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 8.                                                          | Details of MCC                                                                                              | IN-SPACE authorized MCC, owned by Host Entity located at:<br>TM2Space Technologies Private Limited,<br>IIIT-Hyderabad Vindhya C4, Professor CR Road,<br>Gachibowli, Telangana- 500032<br>(IN-SPACE Authorization Ref.: PMA/IN-SPACE/AUTH/2025/108 dated 17 Nov. 2025.)                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 9.                                                          | Master Registry no.                                                                                         | INRSO/DOS/SC/2025/011-01 (Hosted Payload) dated. 29 Dec. 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 10.                                                         | Frequencies Used                                                                                            | <p>The SBB-1 Hosted Payload is totally dependent on the communication channel and MCC of the Host Entity (TM2Space).</p> <p>The Uplink is primarily proposed through the S band communication channel with the UHF channel as secondary/backup. Freq. details are as below:</p> <table><thead><tr><th>Telemetry</th><th>Telecommand</th><th>Data Uplink</th><th>Data Downlink</th></tr></thead><tbody><tr><td>1. 401.5 MHz (UHF)<br/>Frequency band: 401.4928–401.5072 MHz</td><td>1. 2062 MHz<br/>Frequency band: 2059 – 2065 MHz<br/><br/>2. 401.5 MHz<br/>Frequency band: 401.4928–401.5072 MHz</td><td>No data uplink needed.</td><td>No separate data downlink, same telemetry channel is used.</td></tr></tbody></table> | Telemetry                                                  | Telecommand | Data Uplink | Data Downlink | 1. 401.5 MHz (UHF)<br>Frequency band: 401.4928–401.5072 MHz | 1. 2062 MHz<br>Frequency band: 2059 – 2065 MHz<br><br>2. 401.5 MHz<br>Frequency band: 401.4928–401.5072 MHz | No data uplink needed. | No separate data downlink, same telemetry channel is used. |
| Telemetry                                                   | Telecommand                                                                                                 | Data Uplink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Data Downlink                                              |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 1. 401.5 MHz (UHF)<br>Frequency band: 401.4928–401.5072 MHz | 1. 2062 MHz<br>Frequency band: 2059 – 2065 MHz<br><br>2. 401.5 MHz<br>Frequency band: 401.4928–401.5072 MHz | No data uplink needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | No separate data downlink, same telemetry channel is used. |             |             |               |                                                             |                                                                                                             |                        |                                                            |
| 11.                                                         | Security Mechanism to prevent unauthorized access to the Payload by                                         | 1. Dedicated Microcontroller with Encrypted Data Handling All sensor data from BME280, BNO055, and PT100 RTD is processed and encrypted onboard using a secure microcontroller before being transmitted over RS485.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                            |             |             |               |                                                             |                                                                                                             |                        |                                                            |



**Exhibit-2**  
**Technical Parameters**

| Sl. No. | Category                      | Parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Mission Objective             | For educational content for school students by monitoring the environment and orientation in Low Earth Orbit using compact, reliable, and space-suitable sensors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.      | Hosted Payload Name           | Students of BlueBlocks-1 (SBB-1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.      | Hosted Payload Specifications | <ul style="list-style-type: none"> <li>• Payload dimensions – 90 x 90 x 35 mm</li> <li>• Dry and Wet Mass: ~0.1 kg</li> <li>• No. of Sensors: Three (3)               <ol style="list-style-type: none"> <li>1. BME280 (Environmental Sensor):<br/>Function: Measures atmospheric pressure, temperature, and humidity. Pressure Range: 300–1100 hPa<br/>Temperature Range: -40°C to +85°C<br/>Humidity Range: 0–100% RH Interface: I2C/SPI</li> <li>2. BNO055 (9-axis Absolute Orientation Sensor):<br/>Function: Provides orientation data using onboard sensor fusion of accelerometer, gyroscope, and magnetometer.<br/>Output: Euler Angles, Quaternion, Linear Acceleration<br/>Accuracy: Orientation accuracy up to <math>\pm 1^\circ</math> Interface: I2C</li> <li>3. 206 PT RTD (Precision Temperature Sensor):<br/>Type: Platinum RTD (typically PT100 or PT1000)<br/>Temperature Range: -55°C to +150°C<br/>Accuracy: <math>\pm 0.1^\circ\text{C}</math> (depending on conditioning circuit)<br/>Form Factor: 1206 SMD for compact integration</li> </ol> </li> </ul> <p>Communication System:</p> <ul style="list-style-type: none"> <li>○ Type: RS485 (differential serial communication)</li> <li>○ Purpose: Robust communication for transmitting sensor data to Onboard Computer (OBC) or external modules.</li> <li>○ Advantages: Noise immunity, long distance data integrity, multiple node support.</li> </ul> <p>Data Communication Channel Details:</p> <ul style="list-style-type: none"> <li>○ Communication Channel Ownership: The communication link to ground (data receive stations) is provided by the Host Space Object.</li> <li>○ The Hosted Payload interfaces with the Host via an internal RS485 connection.</li> </ul> <p>Onboard Interface (Payload to Host Space Object):</p> <ul style="list-style-type: none"> <li>○ Type: RS485 (differential serial interface)</li> <li>○ Data Rate: Up to 1 Mbps (configured typically around 9600–115200 bps for reliability) Protocol: Custom/standard serial protocol for sensor data packets</li> </ul> |



|                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the Host Entity as well as the other payloads owners hosted in the Host Space Object</p> | <p>2. End-to-End Data Encryption Payload data is encrypted using AES 256 or equivalent secure protocol at the source. Only authorized ground stations with matching decryption keys can access raw data.</p> <p>3. Isolated Communication Channel RS485 channel is physically and logically isolated from other payloads. Custom protocol frames and address filtering ensure that data is only accessible to intended recipients.</p> <p>4. Authentication Protocols All command and telemetry interfaces support mutual authentication, ensuring only verified commands from the host space object are accepted.</p> <p>5. Tamper Detection The system monitors for anomalies in command patterns or access attempts. Upon detection, it switches to a secure mode and logs the event.</p> <p>6. Access Control Layers Software-level partitioning and hardware firewalls restrict internal access to payload data, preventing Host or co payloads from tapping into sensitive information.</p> |
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